Mark schemes

Q1.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$

Accept p = 0.5

B1

1

(ii) $\int \sqrt{x}\, dx = \frac{x^{1.5}}{1.5} \{+c\}$

Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) Area = $\int_1^4 \sqrt{x}\, dx$

Limits 1 and 4 PI

B1

........= $\frac{4^{1.5}}{1.5} - \frac{1}{1.5}$

F(4) – F(1)

M1

= $\frac{14}{3}$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$

Index (p – 1) ft

M1

When x = 4, y' (4) = 0.25

Attempt to find y'(4)

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4}(x - 4)$

accept other forms

A1

4

(ii) When $x = 0$, $y = 1$ B (0, 1)

Subs x= 0 ***and*** *then y = 0 into equation of tangent. PI*

M1

When $y = 0$, $x = -4$ $A(-4, 0)$

Correct ft y_B and x_A
(may be awarded as part of area calculation)

A1ft

Area = 0.5(1)(4) = 2

ft wrong sloping tangent and max of 1 further slip. Final answer must be +'ve

A1ft

3

(c) Translation

'Translation'/'translate(d)'

B1

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Accept equivalent in words provided linked to 'translation/move/shift'
(Note:B0B1 is possible)

B1

2

(d) $h = 1$

PI

B1

Integral = $h/2$ {…….}
{….} = f(1) + 2[f(2) + f(3)] + f(4)
OE summing of areas of the three traps

M1

{….} = 0 + 2(1 + $\sqrt{2}$) + $\sqrt{3}$
Condone 1 numerical slip

A1

Integral = $\frac{1}{2}$ {2 (1 + 1.414…) + 1.732}

Integral = 0.5 × 6.560… = 3.28 to 3sf
Must be 3.28

A1CAO 4

[19]

Q2.

(a) $\frac{1}{4}e^{4x}$

B1 1

(b) $\int e^{4x}(2x + 1)\,dx$
$u = 2x + 1$ $\quad$ $dv = e^{4x}$
by parts

M1

$du = 2$ $\quad$ $v = \frac{1}{4}e^{4x}$

$= \frac{1}{4}(2x + 1)e^{4x} - \frac{1}{2}\int e^{4x}\,dx$
Their (uv – ∫ vdu)

M1

$= \frac{1}{4}(2x + 1)e^{4x} - \frac{1}{8}e^{4x}\ (+c)$

A1 3

(c) $u = 1 + \ln x$

$\frac{du}{dx} = \frac{1}{x}$ or $\frac{dx}{du} = e^{u-1}$

B1

$\int = \int u \, du = \frac{u^2}{2}$ (+c)

in terms of u only

M1A1

$= \frac{(1+\ln x)^2}{2} (+c)$

A1

4

[8]

Q3.

(a) Stretch (parallel) to x-axis

B1

Scale factor $\frac{1}{2}$

B1

Translate $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

B1,B1

4

(b) $A = \int e^{2x} + 3 \, dx$

(+ attempt to integrate)

M1

$= \left[\frac{1}{2}e^{2x} + 3x\right]$

(correct)

A1

$\left(\frac{1}{2}e^{8} + 12\right) - \left(\frac{1}{2}e^{4} + 6\right)$

Substitute 2,4 into their ∫

m1

$\frac{1}{2}(e^8 - e^4) + 6$

$\left(\frac{1}{2}e^4(e^4 - 1) + 6\right)$

A1

4

(c) $x_1 = 2$, $y_1 = e^4 + 3$ (57.6)

Attempt at y(2) or y(4)

M1

$x_2 = 4$, $y_2 = e^8 + 3$ (2980)

Both correct

A1

Area of $A + B =$
$2(e^8 - e^4) + 2(e^8 + 3)$

Attempt to find correct area

M1

Area $B =$
$4e^8 - 2e^4 + 6$
$-\frac{1}{2}e^8 + \frac{1}{2}e^4 - 6$
$= \frac{7}{2}e^8 - \frac{3}{2}e^4$

A1

4

[12]

Q4.

(a) $3x - 5 = A(2x - 1) + B(x + 3)$

M1

$x = -3$ $A = 2$, $x = \frac{1}{2}$ $B = -1$

m1: sub in 2 values or set up simultaneous equations m1A1

3

(b) $\int\left(\frac{2}{x+3}-\frac{1}{2x-1}\right)dx$

M1

$= 2 \ln (x + 3) - \frac{1}{2} \ln (2x - 1)(+c)$

Use of lns for either integral

m1

ft A and B

A1ft

3

[6]

Q5.

(a) $\sin 2x = 2 \sin x \cos x$

B1

1

(b) (i) $A = B = x$

M1

$\cos 2x = \cos^2 x - \sin^2 x$

A1

2

(ii) $\cos (2x + x) = \cos 2x \cos x - \sin 2x \sin x$

M1

$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin^2 x \cos x$

ft (a) and (b)(i)

A1ft

$= \cos^3 x - 3 (1 - \cos^2 x) \cos x$

Eliminate all sines

M1

$= 4 \cos^3 x - 3 \cos x$

AG; convincingly obtained

A1

4

(c) $\cos^3 x = \frac{1}{4}(\cos 3x + 3\cos x)$

Ignore middle sign

M1

$$\int_0^{\frac{\pi}{2}} \cos^3 x \, dx = \frac{1}{4}\left[\frac{1}{3}\sin 3x + 3\sin x\right]_0^{\frac{\pi}{2}}$$

1 for integraiting each term

A1A1

Use of limits

M1

$$\frac{1}{4}\left(-\frac{1}{3}+3\right) = \frac{2}{3}$$

AG

A1

5

[12]

Q6.

(a) $v = \int a \, dt$

$= (20t^2 + t^3)\mathbf{i} - 5e^{-4t}\,\mathbf{j} + \mathbf{c}$

M1 for either term correct.
*Condone no '+ **c**'.*

M1A1

When $t = 1$,
$6\mathbf{i} - 5e^{-4}\mathbf{j} = 21\mathbf{i} - 5e^{-4}\mathbf{j} + \mathbf{c}$

*Finding '+ **c**'; not using **c** = 6**i** – 5e^{-4} **j**.*

M1

$\mathbf{c} = -15\mathbf{i}$

A1

$\mathbf{v} = (20t^2 + t^3 - 15)\mathbf{i}\ 5e^{-4t}\,\mathbf{j}$

A1

5

(b) When $t = 0$, $\mathbf{v} = -15\mathbf{i} - 5\mathbf{j}$

M1

Speed is $\sqrt{15^2 + 5^2}$

M1

$= 15.8 \text{ m s}^{-1}$

Accept $5\sqrt{10}$

A1

3

[8]

Q7.

(a) Using $F = ma$

$1600 \dfrac{dv}{dt} = 4000 - 40v$

$\dfrac{dv}{dt} = \dfrac{4000 - 40v}{1600}$

M1

$\dfrac{dv}{dt} = \dfrac{100 - v}{40}$

A1

2

(b) $40 \dfrac{dv}{100 - v} = dt$

B1

$40 \displaystyle\int \frac{dv}{100 - v} = \int dt$

M1

$-40 \ln(100 - v) = t + c$

Condone lack of '+ c'

A1

When $t = 0$, $v = 0 \Rightarrow c = -40 \ln 100$

$-40 \ln(100 - v) = t - 40 \ln 100$

$t = 40 \ln \frac{100}{100-v}$

$e^{\frac{t}{40}} = \frac{100}{100-v}$

M1A1

$v = 100 - 100\, e^{\frac{t}{40}}$ or $100(1 - e^{\frac{t}{40}})$

A1

6

[8]

Q8.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$\mathbf{a} = (6t - 1.2\,t^2)\,\mathbf{i} + 2\,e^{-2t}\,\mathbf{j}$

A1

2

(b) $v = \int \mathbf{a}\, dt$

$= (3\,t^2 - 0.4\,t^3)\,\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$

when $t = 0$, $\mathbf{r} = 7\,\mathbf{i} - 4\,\mathbf{j}$

condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\,\mathbf{i} - 3\mathbf{j}$

$\mathbf{v} = (7 + 3\,t^2 - 0.4\,t^3)\,\mathbf{i} - (3 + e^{-2t})\,\mathbf{j}$

*ft from ke^{-2t} in (b); just adding 7**i** – 4**j**, m0 accept unsimplified. CAO*

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\,\mathbf{i} - 3.135\,\mathbf{j}$

ft from (b)

M1A1

speed = $\sqrt{9.6^2 + 3.135^2}$

m1

= 10.1 ms $^{-1}$

ft from (b)

A1

4

[10]

Q9.

(a) using $F = ma$

$0.4\ \frac{dv}{dt} = 2 - 4v$

M1

$\frac{dv}{dt} = -10\ (v - 0.5)$

Needs line above

A1

2

(b) hence $\int \frac{1}{v - 0.5} dv = -\int 10\ dt$

$\ln(v - 0.5) = -10t + c$

M1 for any side integrated correctly

M1A1

m1 for + c (and M1 gained)

m1

$v - 0.5 = Ce^{-10t}$

$t = 0,\ v = 1$

$\therefore C = 0.5$

A1

$\therefore v = 0.5 + 0.5e^{-10t}$

condone $v = 0.5 + e^{-10t-0.693}$

A1

5

(c) when $v = 0.55$, $0.55 = 0.5 + 0.5e^{-10t}$

substitute 0.55 into C's (b), after finding c, possible numerical error

M1

$10 = e^{10t}$
$t = \ln 10 \div 10$

A1

$= 0.230$

A1

3

[10]

Q10.

(a) (i) $a = \dfrac{dv}{dt}$

$= 12t + 8e^{-4t}$ m s^{-2}

M1 for either term correct

M1A1

2

(ii) When $t = 0.5$, $a = 6 + 8 \times e^{-2}$

m1

$= 7.08$ m s^{-2}

Condone 7.07

SC1 for 7.1 with no working

A1

2

(b) Using $F = ma$:
$F = 4 \times 7.08$
$= 28.3$ N

Ft from value awarded A1

B1ft

1

(c) $r = \int v\,dt$

At least two terms correct

M1

$= 2t^3 + \dfrac{1}{2}e^{-4t} + 8t + c$

Does not need + c

A1

When $t = 0$, $r = 0 \rightarrow c = -\frac{1}{2}$

Does not need $c = -\frac{1}{2}$

m1

$r = 2t^3 + \frac{1}{2}e^{-4t} + 8t - \frac{1}{2}$

Need r, s (or words)

A1 4

[9]

Q11.

(a) (i) $F = 65g - 260v$

Accept $260v - 65g$

$= 65(9.8 - 4v)$

AG must see $65v$ or 260

B1 1

(ii) Using $F = ma$

$65\frac{dv}{dt} = 65(9.8 - 4v)$

Need to see terms in m (condone – sign)

M1

$\frac{dv}{dt} = -4(v - 2.45)$

AG

A1 2

(b) $\frac{1}{v - 2.45}\frac{dv}{dt} = -4$

B1

$$\int \frac{1}{v-2.45}\,dv = -\int 4\,dt$$

$\ln(v - 2.45) = -4t + c$

M1: log side correct

M1

$-4t + c$

A1

$v - 2.45 = Ce^{-4t}$

$t = 0,\ v = 19.6$

$\therefore C = 17.15$ or $e^{2.84}$

Or $c = \ln 17.15$ or 2.84

A1

$\therefore v = 2.45 + 17.15e^{-4t}\ \ 2.45 + 17.2e^{-4t}$

A1

5

[8]

Q12.

(a) $\mathbf{a} = \dfrac{dv}{dt}$

$\mathbf{a} = -8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}$

*M1: Differentiating with either of the two components correct. Do not need to see **i** or **j**.*

*A1: Correct **i** component.*

*A1: Correct **j** component.*

M1
A1
A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$

$\mathbf{F} = 5 \times \{-8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}\}$

$= -40e^{-2t}\,\mathbf{i} + (30 - 30t)\mathbf{j}$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1
A1
2

(ii) Magnitude of $\mathbf{F}$ is

$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$
M1

$= 50$

A1: Correct answer only.
In this part, condone lack of negative signs in expression for force in (b) (i).
A1
2

(c) When $\mathbf{F}$ acts due west, $\mathbf{j}$ component is zero

$30 - 30t = 0$

M1: Putting $\mathbf{j}$ *component equal to zero.*
M1

$t = 1$

A1: Correct time.
A1
2

(d) $\mathbf{r} = -2e^{-2t}\,\mathbf{i} + (3t^2 - t^3)\mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct. Do not need to see $\mathbf{i}$ *or* $\mathbf{j}$*.*
A1: Correct $\mathbf{i}$ *component.*
A1: Correct $\mathbf{j}$ *component.*
Condone lack of + $\mathbf{c}$
M1
A1
A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j}$ $\therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding $\mathbf{c}$ *using* $6\mathbf{i} + 5\mathbf{j}$ *and* $e^0 = 1$*.*
dM1

$\therefore \mathbf{r} = (8 - 2e^{-2t})\mathbf{i} + (5 + \frac{1}{4}t^4 - t^3)\mathbf{j}$

A1: Correct position vector.
A1
5

[14]

Q13.

(a) Using $F = ma$

$-2mv^{\frac{5}{4}} = m\frac{\mathrm{d}v}{\mathrm{d}t}$

$\therefore \frac{\mathrm{d}v}{\mathrm{d}t} = -2v^{\frac{5}{4}}$ **AG**

B1: Must see $-2mv^{\frac{5}{4}} = m\frac{\mathrm{d}v}{\mathrm{d}t}$ *or* $-2mv^{\frac{5}{4}} = ma$ *and correct final answer.*

B1

1

(b) $$\int \frac{\mathrm{d}v}{v^{\frac{5}{4}}} = -2\int \mathrm{d}t$$

M1: Two integrals with one in the form $\int f(v)\mathrm{d}v$ *where* $f(v) = v^{\pm\frac{5}{4}} \text{ or } v^{\pm\frac{4}{5}}$. *The other integral must not contain v terms.*

M1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t + c$$

A1: Correct expression.

Condone lack of + c for this A1, but no subsequent marks if no c.

A1

When $t = 0$, $v = 16 \Rightarrow c = -2$

dM1: Using $t = 0$ and $v = 16$ to find c.

A1: Obtaining c = –2.

dM1

A1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t - 2$$

$$v^{\frac{1}{4}} = \frac{2}{1+t}$$

$$v = \left(\frac{2}{t+1}\right)^4$$ **AG**

A1: Correct final answer. Must see $v^{\frac{1}{4}} = \frac{2}{1+t}$

or $v^{-\frac{1}{4}} = \frac{1+t}{2}$ or $\frac{1}{v^{\frac{1}{4}}} = \frac{1+t}{2}$

Or

if they obtain $v = \left(\frac{2}{t+c}\right)^4$

v = 16, t = 0 $\Rightarrow$ $16^{\frac{1}{4}} = \frac{2}{c}$, *condone* c = 1 *(no other root considered)*

A1

5

[6]

Q14.

(a) Using $F = ma$,

$$-0.2mv^{\frac{1}{2}} = m\frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

AG Must see equ'n containing m

B1

1

(b) $$\int \frac{dv}{v^{\frac{1}{2}}} = -\int 0.2\, dt$$

M1

$$2v^{\frac{1}{2}} = -0.2t + c$$

m1 for + c

A1m1

When $t = 0, v = 16 \therefore C = 8$

A1

$$2v^{\frac{1}{2}} = -0.2t + 8$$

$$v = (4 - 0.1t)^2$$

AG

A1

5

(c) When $v = 1$, $1 = (4 - 0.1t)^2$
$4 - 0.1t = \pm 1$

M1

$t = 30$ or 50

$\left[\text{if use } 2v^{\frac{1}{2}} = 8 - 0.2t \text{ no need to see } 50\right]$

A1

$t = 30$

$t \neq 50$ as ball stops when $t = 40$

A1

3

(d) Integrating $v = (4 - 0.1t)^2$

$v = 16 - 0.8t + 0.01t^2$

$x = 16t - 0.4t^2 + \dfrac{0.01}{3}t^3 + d$

M1 for first 3 terms or $-\dfrac{10}{3}(4 - 01t)^3$

M1

When $t = 0, x = 0 \Rightarrow d = 0$

A1

$x = 16t - 0.4t^2 + \dfrac{0.01}{3}t^3$

When speed is $1\text{ m s}^{-1}, t = 30$

$x = 480 - 360 + 90$

dep on M1 above

m1

= 210

[No 'd', 3 marks only]

A1

4

[13]

Q15.

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{\lambda}{v^{\frac{1}{4}}}$$

M1

$$\int v^{\frac{1}{4}}\,\mathrm{d}v = -\int \lambda\,\mathrm{d}t$$

Condone one of $v^{-\frac{1}{4}}, +\int \lambda \mathrm{d}t, \frac{1}{\lambda}$

m1

$$\frac{4}{5}v^{\frac{5}{4}} = -\lambda t + c$$

m1 for + c

A1A1m1

$$t = 0, v = u \therefore c = \frac{4}{5}u^{\frac{5}{4}}$$

A1

$$\therefore v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t$$

$$v = \left(u^{\frac{5}{4}} - \frac{5}{4}\lambda t\right)^{\frac{4}{5}}$$

A1

[7]

Q16.

$$r = \int v\ \mathrm{d}t$$

M1

= $t^4 + 4\cos 2t + 5t$ (+ c)

A1

When $t = 0$, $r = 0 \Rightarrow c = -4$

Finding c *correctly*

M1

$\therefore\ r = t^4 + 4\cos 2t + 5t - 4$

A1ft

4

[4]

Q17.

(a) Using $F = ma$:
$-0.08v^2 = 0.05a$

B1

$\therefore \frac{dv}{dt} = -1.6v^2$

AG; condone sign error in first B1

B1

2

(b) $\int \frac{dv}{v^2} = -1.6\int dt$

M1

$-\frac{1}{v} = -1.6t(+c)$

Condone $-\frac{1}{v} = -1.6t + c \Rightarrow \frac{1}{v} = 1.6t + c$

A1

When $t = 0, v = 3 \Rightarrow c = -\frac{1}{3}$

M1

$\frac{1}{v} = \frac{1}{3} + 1.6t$ *

A1

$$\frac{1}{v} = \frac{1}{3} + \frac{8}{5}t$$

$$\frac{1}{v} = \frac{5 + 24t}{15}$$

$$v = \frac{15}{5 + 24t}$$

*AG; all working lines correct from **

A1

5

[7]

Q18.

(a) (i) $a = \frac{dv}{dt} = 6t - 6\cos 3t$

M1 for at least one term correct

M1A1

2

(ii) When $t = \frac{\pi}{3}, a = 6 \times \frac{\pi}{3} - 6\cos(3 . \frac{\pi}{3})$

M1

$= 2\pi + 6$

AG

A1

2

(b) $r = t^3 + \frac{2}{3}\cos 3t + 6t + c$

M1 for 3 terms including cos3t term
Condone no '+ c'

M1A1

When $t = 0$, $r = 0$ $\therefore c = -\frac{2}{3}$

M1

$\therefore r = t^3 + \frac{2}{3}\cos 3t + 6t - \frac{2}{3}$

A1

4

[8]

Q19.

(a) Using $F = ma$

$-0.05mv = m\frac{dv}{dt}$

$\therefore \frac{dv}{dt} = -0.05v$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05\,dt$

B1

$\ln v = -0.05t + c$

$v = Ce^{-0.05t}$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$\therefore C = 20$

$v = 20e^{-0.05t}$

only for fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$e^{0.05t} = 2$

A1

$\therefore t = \frac{1}{0.05}\ln 2$

$= 13.9$ or $20\ln 2$

A1

3

(d) Work done by force is change in KE of car

M1

$= \frac{1}{2}m(20)^2 - \frac{1}{2}m(10)^2$

A1

$= 150\ m$

A1

3

[11]

Q20.

(a) Using $F = ma$

$-0.05mv = m\frac{dv}{dt}$

$\therefore \frac{dv}{dt} = -0.05v$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05\ dt$

B1

$\ln v = -0.05t + c$

$v = Ce^{-0.05t}$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$\therefore C = 20$

$v = 20e^{-0.05t}$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$e^{0.05t} = 2$

A1

$\therefore t = \frac{1}{0.05} \ln 2$
$= 13.9$
Accept 20 ln 2

A1
3
[8]

Q21.

(a) Using $F = ma$:
$2400\mathbf{i} - 4800t\mathbf{j} = 800\mathbf{a}$

M1

$\mathbf{a} = 3\mathbf{i} - 6t\mathbf{j}$

A1
2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

M1

$= 3t\mathbf{i} - 3t^2\mathbf{j} + \mathbf{c}$
*Condone no '+ **c**'*

A1

When $t = 0$, $\mathbf{v} = 6\mathbf{i} + 30\mathbf{j}$
$\therefore \mathbf{c} = 6\mathbf{i} + 30\mathbf{j}$
*Needs '+ **c**' above*

M1

$\therefore \mathbf{v} = (3t + 6)\mathbf{i} + (30 - 3t^2)\mathbf{j}$
AG

A1
4

(c) $\mathbf{r} = \int \mathbf{v} \, dt$

M1

$= (\frac{3}{2}t^2 + 6t)\,\mathbf{i} + (30t - t^3)\,\mathbf{j} + \mathbf{d}$
*A1 **i** term, A1 **j** term; condone no '+ **d**'*

A1,A1

When $t = 0$, $\mathbf{r} = 2\mathbf{i} + 5\mathbf{j}$

$\therefore$ $\mathbf{d} = 2\mathbf{i} + 5\mathbf{j}$

M1

$\therefore$ $\mathbf{r} = (\frac{3}{2}t^2 + 6t + 2)\mathbf{i} + (30t - t^3 + 5)\mathbf{j}$

A1

5

[11]

Q22.

(a) Using $F = ma$:

$-\lambda mv = ma = m\frac{dv}{dt}$

Condone no '–'

M1

$\therefore \frac{dv}{dt} = -\lambda v$

AG

Note: no use of m $\Rightarrow$ no marks in (a)

A1

2

(b) $\int \frac{dv}{v} = -\lambda \int dt$

M1

$\ln v = -\lambda t + c$

$v = C e^{-\lambda t}$

Needs '+ c'

A1

When $t = 0$, $v = U \Rightarrow C = U$

Needs correct working

M1

$v = U e^{-\lambda t}$

AG

A1

4

[6]

Q23.

(a) (i) $a = 2 + 12e^{-t}$

Differentiating, with at least one term correct.
Correct velocity

M1A1

2

(ii) $2 < a \leq 14$

For 2, For 14

B1, B1

Correct inequalities

B1

3

(b) $s = t^2 + 12e^{-t} + c$

Integrating, with at least one term correct.

M1

Correct expression with or without c

A1

$s = 0, t = 0 \Rightarrow c = -12$

Finding c

dM1

$s = t^2 + 12e^{-t} - 12$

Correct final expression

A1

4

[9]

Q24.

(a) The boat is not affected by the movement of the water.

First assumption

B1

The resistance force will directly oppose the motion of
the boat and be the only force that needs to be considered.

second assumption

B1

2

(b) The boat is a particle. / There is no wind. / No air resistance.

Appropriate assumption

B1

1

(c) (i) $80\frac{dv}{dt} = -20v$

Use of Newton's second law, $\frac{dv}{dt}$ *and 20v.*

M1

$\frac{dv}{dt} = -\frac{v}{4}$

Correct result

A1

2

(ii) $\int \frac{1}{v}\,dv = -\int \frac{1}{4}\,dt$

Sep. of variables and forming integrals

M1

$\ln v = -\frac{t}{4} + c$

Integrating to get an 1n v term

dM1

Correct integrals with or without c

A1

$v = Ae^{-\frac{t}{4}}$

$v = 12,\ t = 0 \Rightarrow A = 12$

Finding A or c

dM1

$v = Ae^{-\frac{t}{4}}$

Correct final result

A1

5

[10]

Q25.

(a) $20\frac{dv}{dt} = -10\sqrt{v}$

applying Newton's second law with $\frac{dv}{dt}$

M1

correct differential equation

A1

$\frac{dv}{dt} = -\frac{\sqrt{v}}{2}$

separating variables

dM1

$\int \frac{1}{\sqrt{v}}\,dv = \int -\frac{1}{2}\,dt$ AG

$2\sqrt{v} = \frac{t}{2} + c$

integrating

dM1

correct integrals

A1

$t = 0$, $v = 25$ $\Rightarrow$ $c = 10$

finding the constant of integration

$v = \left(\frac{20-t}{4}\right)^2$

correct final result from correct working

A1

7

(b) $t = 20$

correct time

B1

1

[8]

Q26.

(a) $F = 800 + \frac{1200}{20}t = 800 + 60t$

finding the gradient of the line

M1

correct gradient

A1

$1200a = 800 + 60t$

correct intercept

B1

$a = \frac{800}{1200} + \frac{60}{1200}t = \frac{2}{3} + \frac{t}{20}$

using Newton's second law on two terms

dM1

AG

correct result from correct working

A1

5

(b) $v = \int \frac{2}{3} + \frac{t}{20}\,dt = \frac{2t}{3} + \frac{t^2}{40} + c$

integrating

M1

correct integral with or without c

A1

$v = 0,\ t = 0 \Rightarrow c = 0$

A1

$v = \frac{2t}{3} + \frac{t^2}{40}$

showing c = 0

A1

3

(c) $s = \int_0^{20} \frac{2t}{3} + \frac{t^2}{40}\,dt$

integrating

M1

correct integral, with or without c.

A1

$$= \left[\frac{t}{3} + \frac{t^3}{120}\right]_0^{20}$$

use of both limits or finding c

dM1

= 200 m

correct distance

A1

4

(d) The $\frac{2t}{3}$ term would change, because only

correct term

B1

the constant term in the force would change.
When integrated this becomes the t term in the velocity.

correct explanation

B1

2

[14]

Q27.

$$\frac{dv}{dt} = \frac{k}{v}$$

B1

$\int v\,dv = \int k\,dt$

Separation of variables involving t

M1

$$\frac{v^2}{2} = kt + c$$

m1

Integrate

A1

$t = 0$, $v = u$, $\therefore c = \frac{u^2}{2}$

m1

$v^2 = u^2 + 2kt$

A1

[6]

Q28.

(a) $mg - 0.1m\,v = m\frac{\mathrm{d}v}{\mathrm{d}t}$

Newton's 2nd law

M1

$\frac{\mathrm{d}v}{\mathrm{d}t} = g - 0.1v$

CAO

A1 2

(b) $\int_0^v \frac{\mathrm{d}v}{g - 0.1v} = \int_0^t \mathrm{d}t$

Attempt at integration with correct separation of variables

M1

$[-10\ln(g - 0.1v)]_0^v = [t]_0^t$

–1 EE

A2,1

$\ln \frac{g - 0.1v}{g} = -0.1t$

A1F

$v = 10g(1 - e^{-0.1t})$

Or equivalent

A1F 5

(c) As $t \to \infty$, $e^{-0.1t} \to 0$

M1

1

$\therefore \; v \to 10g = 98$

[8]

Q29.

(a) $v = \int t - \frac{t^2}{5}\,dt$

Integrating both terms

M1

$= \frac{t^2}{2} - \frac{t^3}{15} + c$

Correct integral with or without c

A1

$v = 0$, $t = 0 \Rightarrow \; c = 0$

Showing $c = 0$

A1

3

$v = \frac{t^2}{2} - \frac{t^3}{15}$

(b) $v(5) = \frac{5^2}{2} - \frac{5^3}{15} = 4.17\text{ ms}^{-1}$

Substituting $t = 5$

M1

Correct v

A1

2

(c) $s = \int_0^5 \left(\frac{t^2}{2} - \frac{t^3}{15}\right) dt$

Integrating

M1

$= \left[\frac{t^3}{6} - \frac{t^4}{60}\right]_0^5$

Correct expression

A1

Substitution of two limits or finding c and substituting t = 5

m1

= 10.4 m

Correct distance
***sc** for only one limit M1A1A1*

A1

4

[9]

Q30.

(a) $v = \int 20 \sin 4t \, dt$

Attempt to integrate a

M1

$= -5 \cos 4t + c$

Correct integral with or without c

A1

$t = 0$, $v = 0 \Rightarrow c = 5$

Finding c

M1

$v = 5 - 5 \cos 4t$

Correct c

A1

4

(b) $s = \int 5 - 5\cos 4t \, dt$

Attempt to integrate v

M1

$= 5t - \frac{5}{4} \sin 4t + c$

Correct integral with or without c

A1

$t = 0$, $s = 0.8 \Rightarrow c = 0.8$

Finding c

M1

$s = 5t - \frac{5}{4}\sin 4t + 0.8$

Correct c

A1

4

[8]

Q31.

(a) $\mathbf{a} = \frac{1000t\mathbf{i} - 5000\mathbf{j}}{2000} = \frac{t}{2}\mathbf{i} - \frac{5}{2}\mathbf{j}$

Using Newton's 2nd law

M1

Correct acceleration

A1

2

(b) $\mathbf{v} = \left(\frac{t^2}{4} + c\right)\mathbf{i} + \left(\frac{-5t}{2} + d\right)\mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

t = 0, $\mathbf{v} = 6\mathbf{j} \Rightarrow c = 0, d = 6$

Evaluation of constants

m1

Correct constants and correct final answer

A1

4

$\mathbf{v} = \left(\frac{t^2}{4}\right)\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$

(c) $\mathbf{r} = \left(\frac{t^3}{12} + e\right)\mathbf{i} + \left(6t - \frac{5t^2}{4} + f\right)\mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0$, $\mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow e = 0, f = 0$

Evaluation of constants

m1

$\mathbf{r} = \left(\frac{t^3}{12}\right)\mathbf{i} + \left(6t - \frac{5t^2}{4}\right)\mathbf{j}$

Correct constants and correct final answer

A1

4

[10]

Q32.

(a) $v = \int 2 - 2e^{-t}\,dt = 2t + 2e^{-t} + c$

Integrating

M1

$v = 4, t = 0 \Rightarrow c = 2$

Correct, with or without c

A1

$v = 2t + 2e^{-t} + 2$

Correct constant from correct working

A1

3

(b) $s = \int 2t + 2e^{-t} + 2\,dt$

Integrating

M1

$= t^2 - 2e^{-t} + 2t + c$

Correct, with or without c

A1

$s = 0, t = 0 \Rightarrow c = 2$

Finding c

m1

$s = t^2 - 2e^{-t} + 2t + 2$

Correct final answer

A1

4

[7]